NAEEM NAIRO

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RESEARCH INTERESTS

Human–machine interfaces for assistive control (surface EMG, eye gaze); prosthetics and rehabilitation engineering; biomedical instrumentation, signal processing and device validation.

EDUCATION

M.S. Biomedical Engineering, University of New Haven, Tagliatela College of Engineering *2025 – present*

GPA 3.9 · Fulbright Scholar · Coursework: Biomedical Sensors and Instrumentation, Signal Analysis, Biomechanics, Biostatistics, Biomedical Polymers

B.S. Mechanical Engineering (Mechatronics), Polytechnic University

Coursework: Mechanical Design, Control Systems, Mechatronic Systems, Manufacturing Processes, Material Science, Thermodynamics

PUBLICATIONS

M. N. Nairokh. Hands-Free Proportional Control Through Eye Gaze: A Low-Cost Webcam Interface Evaluated on a Ball-and-Beam Task. *Under review*, 2026.

RESEARCH PROJECTS

EyeMotion: Eye-Gaze Proportional Control *University of New Haven*

- Built a webcam-based interface (Python, MediaPipe, OpenCV) that maps iris position to a proportional control signal, removing the need for dedicated eye-tracking hardware.
- Evaluated it on a ball-and-beam task, an unstable control benchmark; manuscript under review.

NeuroGrip: EMG-Controlled Robotic Hand *Independent project*

- Controlled an 11-degree-of-freedom, 3D-printed robotic hand in real time from 8-channel forearm EMG with a neural network trained in PyTorch, driving tendon pulleys and knuckle servos through an Arduino.
- Reduced output flicker with a temporal-consistency term in the training loss; built a kinematic simulator from the CAD assembly to check joint ranges.

GrapGrow: Modular Pediatric Prosthetic Arm *With Yale University students, Fall 2026 – present*

- Developing a 3D-printable prosthetic arm sized from anthropometric data for ages 8–14, with a socket, an extension mechanism that grows with the child, and a quick attach/detach interface for accessories.

Rehabilitation platform *Instrumentation and Sensors Laboratory*

- Adapting a ball-and-beam control platform into a rehabilitation tool, with training strategies for stroke patients with mobility impairments.

TEACHING

Graduate Teaching Assistant, Instrumentation and Sensors Laboratory, University of New Haven *Fall 2025 – present*

- Guide students through LabVIEW and NI ELVIS experiments in EMG signal acquisition and temperature sensing.

INDUSTRY EXPERIENCE

Product Design Engineer, AHL Logics *Nov 2021 – Aug 2025*

- Took mobility products from concept through prototyping, testing and mass production; co-inventor, as part of the design team, on 5 U.S. patents pending for a folding mechanism.
- Used structural FEA to cut part weight by more than 15% and validated predictions against physical prototype tests; wrote design specifications, test plans and reports.

SKILLS

Signals and computing: Python, PyTorch, MATLAB, LabVIEW, MediaPipe, OpenCV, statistical analysis (Python, R)

Instrumentation and embedded: EMG acquisition and processing, NI ELVIS, Arduino, ESP32, STM32, I2C

Mechanical design and analysis: SOLIDWORKS (CSWE), ANSYS, structural FEA, DOE, GD&T, 3D printing

Verification: test planning, design verification and validation, FMEA, root-cause analysis

HONORS AND SERVICE

Fulbright Scholar, U.S. government international scholarship (2025–2027) · SOLIDWORKS Champion (2026) · Secretary, Biomedical Engineering Society, University of New Haven (2026–) · President, Mechanical Engineering Society, Polytechnic University (2021) · Member: IEEE, ASME, BMES

CERTIFICATIONS

SOLIDWORKS Design Expert (CSWE) · SOLIDWORKS Simulation Professional · SOLIDWORKS CAD Design Professional · SOLIDWORKS Additive Manufacturing Associate · Additive Manufacturing: Materials for 3D Printing